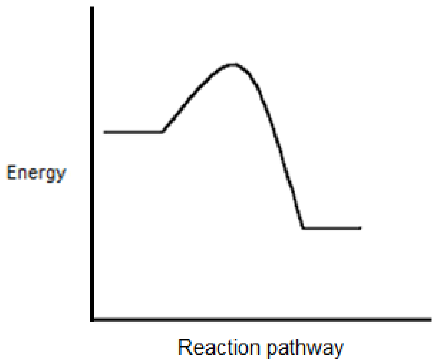
Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
4	(a)	(i)		**Agree** because we are not told / do not know that the carbonate is in excess (1) there may not be enough to neutralise all of the acid (1) OR **Disagree** because the carbonate will be in excess (1) so all of the acid will all be used up / neutralised (1) **no credit** if no choice is made and the answer does not mention agreeing or disagreeing	2			2		2
		(ii)		$K_2CO_3 + 2HCl \rightarrow 2KCl + CO_2 + H_2O$ reactants (1) products (1) balancing (1) balancing mark can only be awarded if both the reactants and products are correct		3		3	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		will overshoot / go past the point of neutralisation / endpoint 'too much acid' is a neutral answer	1			1		1
		(ii)		$H^+ + OH^- \rightarrow H_2O$ (1) charges on ions **must** be present H^+ comes from the (sulfuric) acid and OH^- comes from the (potassium) hydroxide / alkali (1) 'one comes from the acid and the other from the alkali' and 'they come from the acid and alkali' are neutral answers	2			2		
		(iii)			1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		both will give a lilac flame / same colour flame (because both contain the potassium ions / K^+ / potassium)			1	1		1
		(ii)		add silver nitrate solution / $AgNO_3(aq)$ (1) potassium chloride gives a (white) **precipitate** whereas potassium sulfate gives no precipitate / only the chloride gives a (white) **precipitate** (1) OR add barium chloride solution / $BaCl_2(aq)$ (1) potassium sulfate gives a (white) **precipitate** whereas potassium chloride gives no precipitate / only the sulfate gives a (white) **precipitate** (1)		1	1	2		2
				Question 4 total	**6**	**4**	**2**	**12**	**1**	**6**